Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Led a production **tool-calling multi-agent system** on **OpenAI Agents SDK**, **LangGraph**, and **MCP** serving real users across **millions of daily transactions** at 99%+ uptime.
- Built **evaluation harnesses** measuring output quality, **hallucination rates**, and **answer grounding** across agent versions, and gated releases with **A/B tests**.
- Trained and **fine-tuned PyTorch models** on **multi-GPU clusters**, using **quantization** to serve them efficiently in production inference.
- Designed agent workflows across **open-weight and hosted LLMs** (Llama, Gemini, GPT, Claude), grounding multi-hop answers with **GraphRAG**.

QureAI | *AI Engineer* Mar 2026 – Present

- Built **computer vision** models for **radiology images** on **CNN and ResNet** backbones with **UNet segmentation** in **PyTorch**.
- Built an **RCA incident agent** that escalates to a **SQL agent over A2A**, with **Langfuse** tracing and evaluation on production agent runs.
- Cut **p99 inference latency from 420ms to 140ms** across **50K+ daily requests** by moving model serving to async **vLLM** with autoscaling.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Improved vision model accuracy **9.6 points** and cut false negatives **31%** by evaluating **CNN, ResNet, UNet, and ViT** architectures and tuning **augmentation**, **class balancing**, and thresholds.
- Cut model-related production regressions **28%** and validation time from **6 hours to 90 minutes** with an automated **model-evaluation and deployment-gating** framework.
- Raised inference throughput **2.4x** and cut **GPU memory 27%** by profiling and optimizing batching, preprocessing, and concurrency in Python/C++ model services.
- Raised ML service availability to **99.9%** and cut **p99 latency 32%** with async serving, batching, health checks, and automated failure recovery.

Projects

HoG-ViT (M.S. Thesis) `github.com/ganeshhgupta/HoG-ViT`

- **Vision Transformer** that folds **Histogram of Oriented Gradients** features into **patch embeddings** for image classification, trained and evaluated in **PyTorch**.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system that triages production alerts, routes each to a specialized sub-agent with **tool use**, and writes a structured **root-cause brief**.

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- **LLM evaluation** platform with rubric-based scoring, hallucination detection, and regression alerting across model versions; **FastAPI**, React/TypeScript, **PostgreSQL**.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages: Python, SQL, C++, TypeScript

ML and Vision: PyTorch, CNN, ResNet, UNet, ViT, Segmentation, Fine-tuning, Quantization, Reinforcement Learning

LLMs and Agents: OpenAI Agents SDK, LangGraph, MCP, A2A, RAG, GraphRAG, vLLM, TensorRT-LLM, Langfuse, LLM Evals

Infrastructure: FastAPI, PostgreSQL, Docker, Kubernetes, AWS (SageMaker, Bedrock), GCP, Multi-GPU Training, Slurm

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer